

MUQ HUSSAIN

RF / EDGE AI / EMBEDDED LINUX / SPACE SYSTEMS

Toronto, ON | muqti123@gmail.com | linkedin.com/in/muqtadir66 | github.com/muqtadir66 | Portfolio

Engineering Physics and Management graduate specializing in intelligent radio, edge AI, embedded Linux, and real-time technical software. Created a multi-model RF inference orchestrator, leads ESA OPS-SAT PRETTY experimentation, and develops operator, runtime, and acquisition systems across CPU, NPU, and TPU targets.

EXPERIENCE

QOHERENT | RF & Edge AI Systems Developer / Technical Project Lead

2026 - PRESENT

Toronto, Ontario

- Created and technically owns PRISM, a C++ multi-model RF inference orchestrator with lock-free IQ fan-out, real-time telemetry, model lifecycle controls, SigMF capture, SDR control, and headless sweep automation; latest verified suite: 127 tests passing.
- Leads Qoherent's ESA OPS-SAT PRETTY experimentation, characterizing an ARMv7/musl target, cross-building the inference stack, validating NCNN conversions against ONNX outputs, and defining a provenance-tracked POSIX flight-package workflow.
- Led architecture of RIASM, a browser and terminal RF-monitoring application with WebGL spectrum and waterfall views, multi-engine telemetry, scan automation, ML overlays, capture management, alerts, and radio control.
- Extended the founder-created RIACPP engine with Coral, Qualcomm QNN/HTP, TFLite, and NCNN paths plus live-IQ benchmarking; initiated iq-bridge for native libiio acquisition, ZMQ/UDP transport, control, and SigMF recording on constrained Linux systems.

APPLIED PRECISION INC. | Precision Metrology & Engineering Intern

SEP - DEC 2022

Vaughan, Ontario

- Operated Leica RTC 360, Zeiss Comet L3D, and Faro scanning systems for industrial inspection and reverse engineering across automotive, aerospace, plastics, medical-device, and tooling contexts.
- Registered point clouds and produced watertight meshes and as-built/hybrid CAD using Geomagic Design X, PolyWorks, Autodesk Inventor, and SolidWorks.

SELECTED ENGINEERING FOUNDATION

Autonomous Blackjack Dealer: Co-developed a 3D-printed motorized dispenser with Arduino/Raspberry Pi control, Python game logic, MediaPipe gestures, YOLOv8 card recognition, and a Tkinter interface.

CANDU Fuel & Reactor Studies: Modelled fuel configurations in MCNP and analyzed neutron flux, energy deposition, power distribution, xenon transients, and reactor operation through laboratory and simulator work.

PERL Silicon Solar Cell: Fabricated a cell through cleanroom processing, modelled the device in Sentaurus TCAD, characterized electrical performance, and analyzed measured-to-simulated differences.

EDUCATION

McMaster University | B.Eng.Mgt., Engineering Physics & Management

APR 2025

Nuclear Engineering specialization | Hamilton, Ontario

TECHNICAL CORE

C++17/20, Python, CMake, concurrency, embedded Linux, cross-compilation, SDR/IQ, libiio, UHD, GNU Radio, ZMQ/UDP, SigMF, ONNX Runtime, NCNN, QNN/HTP, TFLite, Coral, FastAPI, WebSockets, WebGL2, Git, testing, model validation, Sigil, technical leadership